Architecture Selection and Justification

Self-Service Coffee Kiosk System | Aarya Prasad G | PES1UG24AM331

Architecture selection

We chose a layered architecture for the Self-Service Coffee Kiosk System. It separates the presentation layer (touchscreen UI), the business and hardware-integration responsibilities (Order Manager, Payment Service and Receipt Printer Adapter), and the data layer (Menu & Pricing Database).

Reason 1 - Clear responsibilities

The Touchscreen UI collects the customer's coffee and size choices, while the Order Manager validates the selection and coordinates checkout. Keeping those responsibilities separate makes each part easier to test and maintain when the menu or screen flow changes.

Reason 2 - Fits the kiosk workflow and hardware

The kiosk has a small, defined flow: choose Espresso, Americano or Latte; choose Small or Large; pay by credit card; and receive a printed receipt. A layered design is straightforward for this scope and provides clear integration points for payment processing and the printer without the deployment and monitoring overhead of multiple microservices.

Security advantage

The Payment Service is accessed through a defined interface rather than being embedded in the touchscreen logic. Payment requests should use the approved payment provider's secure interface. The kiosk should retain only a transaction reference and payment status, not raw card details, and secrets must not be hard-coded in source files.

Performance benefit

The Order Manager can retrieve menu and price data through one defined data interface, keeping the normal selection flow simple and avoiding unnecessary network hops between separately deployed services. Payment authorization still depends on the payment provider and network response time.

Components and interfaces shown

Components: Touchscreen UI, Order Manager, Payment Service, Receipt Printer Adapter, and Menu & Pricing Database. Interfaces: OrderControl (order request/status), CardPayment (authorization/result), ReceiptPrint (receipt data), and MenuCatalog (menu and pricing lookup). The diagram uses provided-interface balls and required-interface sockets on the component boundaries.